

UpLift | Predicting Transit Accessibility Failures Before They Strand a Rider

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Motivation

Here is something that happened to me.

I am a Disabled transit rider in Philadelphia. I depend on SEPTA elevators to reach platforms I cannot get to by stairs. On more than one occasion, I have shown up at a station — sometimes running late, sometimes in the rain — and found the elevator out of service. No warning, no reroute. Just a sign taped to a gate.

That is not a Philadelphia problem. It is not even an American problem.

According to the Bureau of Transportation Statistics, 3.6 million Americans with disabilities that affect travel do not leave their homes. [1] Broken accessibility equipment is a direct contributor. The MTA in New York City has elevators at only 23% of its subway stations, and those elevators break down regularly — which means that on any given day, the number of stations a wheelchair user can actually reach is even smaller than that figure suggests. [2]

A 2024 international review covering 34 studies across multiple countries found that physical infrastructure failure — broken lifts, blocked ramps, inaccessible station entrances — is one of the most consistently reported barriers to transit use for Disabled people, across every region studied. [5]

Right now, transit agencies fix elevators and escalators after they break. A maintenance team responds to a failure report. By then, a Disabled rider has already been stranded.

UpLift changes that. It predicts which elevators, escalators, and platform lifts are likely to fail in the next 30 days — so maintenance teams can act before the breakdown happens, not after.

Disabled riders are the primary people this protects, because we have no alternate route when a lift fails. But the tool works for everyone who cannot use stairs: seniors, parents with strollers, people recovering from surgery, travelers with heavy luggage. Every transit agency on earth manages this problem. That is the market.

What UpLift Does

UpLift is a machine learning model that learns from historical maintenance and outage records. Each piece of accessibility equipment — every elevator, escalator, and platform lift in a transit system — gets a score. That score is the model's best estimate of how likely that piece of equipment is to break down unexpectedly in the next 30 days.

The transit agency gets a ranked list: here are the ten units most at risk this month, sorted by failure probability. Maintenance teams use that list to schedule proactive service calls. Riders can optionally receive alerts through existing transit apps.

Think of it like a credit score, but for mechanical equipment. A credit score synthesizes dozens of financial signals — payment history, utilization, age of accounts — into one number that predicts the likelihood of default. UpLift does the same thing with maintenance signals: equipment age, usage load, weather exposure, time since last service. The output is one number per unit. The maintenance team decides what to do with it.

This works because the answer to “will this elevator break” is a yes or no question with a historical record. That makes it a supervised learning classification problem: the model learns from past cases where we know what happened, and applies those patterns to predict what is about to happen.

AI Blueprint

Blueprint Summary

Problem Statement

Elevator, escalator, and platform lift failures strand Disabled transit riders without warning. Agencies respond reactively — after the breakdown. This exposes them to ADA legal liability and causes real harm to real people. Can a machine learning model predict which equipment is likely to fail in the next 30 days, so maintenance happens before a rider is stranded?

User: Transit maintenance and operations teams at agencies worldwide — MTA NYC, SEPTA Philadelphia, CTA Chicago, WMATA DC, London Underground, Tokyo Metro, Delhi Metro, São Paulo Metro, and others.
Secondary user: Disabled riders, via real-time alerts through existing transit apps.

Functionality

Functionality

For each piece of accessibility equipment in a transit system, UpLift produces a probability score (0 to 1) representing the likelihood of an unplanned outage in the next 30 days. Maintenance teams receive a ranked list of at-risk units and use it to schedule proactive service.

How Input → Output Creates Functionality

The model trains on historical records: what was true about a piece of equipment in the months before it failed? At prediction time, it looks at the same variables for every active unit and asks: does this equipment look like equipment that was about to break? The output is a probability score. A score above the chosen threshold triggers a maintenance alert. The threshold can be adjusted by each agency based on their budget and risk tolerance.

Inputs (X) and Label (Y)

Inputs X — what the model learns from

These are the variables the model uses to make its prediction. All twelve are universally available — every transit system on earth tracks them in some form.

- Equipment type (elevator, escalator, platform lift)
- Equipment age in years
- Manufacturer and model
- Days since last scheduled maintenance
- Number of unplanned outages in prior 12 months
- Total usage load (estimated daily trips × days in service)
- Days since last parts replacement
- Most recent inspection score
- Average monthly temperature at that station (from NOAA weather data)
- Station average daily ridership
- Number of floors served
- Whether the station is a major transfer hub (yes/no)

Label Y — what the model is trying to predict

Binary: 1 = this equipment had an unplanned outage within the next 30 days. 0 = it did not.

Labels come from official agency outage logs, matched to equipment ID by timestamp. Each training example represents one piece of equipment over one 30-day window. We roll that window forward month by month to build up as many training examples as the historical record allows.

Conceptual summary:

Input (conceptual): The maintenance and operational profile of one piece of equipment over the past several months.

Output (conceptual): How likely is this equipment to break down unexpectedly in the next 30 days?

Dataset Construction

Dataset Construction (where do the instances come from?)

A note on data availability:

MTA NYC is the only US transit agency with a deep public historical record suitable for model training. Most other agencies publish real-time outage feeds only — not historical archives. This is an important distinction: real-time feeds can be scraped and archived going forward to build training data,

but they cannot substitute for years of labeled historical records. The data strategy below reflects that reality honestly.

Primary training dataset:

MTA NYC Subway Elevator and Escalator Availability data, publicly available at [data.ny.gov](https://data.ny.gov/dataset-identification/rc78-7x78) (dataset ID: rc78-7x78), covering monthly availability metrics going back to 2015 — the longest public historical record of any US transit system. The MTA also publishes real-time outage feeds via its developer API with timestamped equipment IDs, which generate the Y labels. Two additional MTA datasets extend the training corpus: MTA LIRR Elevator and Escalator Availability (beginning 2019, dataset ID: 9hjt-526f) and MTA Metro-North Elevator and Escalator Availability (beginning 2021, dataset ID: ax67-8386). These cover different systems within the same agency and allow the model to test generalization across equipment types and ridership patterns within one data infrastructure. [3, 9, 10]

Weather enrichment:

Temperature and precipitation for each station location, pulled from the NOAA Climate Data Online API matched by GPS coordinates. Weather is a meaningful predictor of equipment stress, particularly for outdoor and partially outdoor stations.

Real-time feeds available for future archiving:

WMATA (DC Metro) publishes a real-time Elevator and Escalator Incidents API returning outage records by station, including symptom type and timestamps. [11] A third-party civic project called DC Metro Metrics scraped and archived this feed from June 2013 to February 2015, publishing the dataset on GitHub — this is usable as a limited supplemental dataset. [12] SEPTA (Philadelphia) publishes a real-time elevator outage API via OpenDataPhilly with no official historical archive. [13] However, civic hackers Ather Sharif and James Tyack began recording SEPTA's real-time feed in 2014 through a tool called Unlock Philly, producing a 2017–2019 analysis covering 31 of Philadelphia's 108 wheelchair-accessible stations — with 56th Street Station on the Market Frankford Line recording the highest outage count. [14] Chicago CTA publishes a real-time Customer Alerts API that includes elevator and accessibility alerts. [15] Transport for London (TfL) publishes lift and escalator status via its Unified API. [16] None of these agencies currently offer official historical breakdown archives, but all can be scraped going forward to build training data under the consortium model described in the scaling section.

National Transit Database (NTD):

The US Department of Transportation publishes 2022–2024 NTD Annual Data on vehicle and equipment breakdowns, covering transit providers nationwide. [17] This dataset is not elevator-specific but provides a national maintenance failure baseline useful for the equity audit and for contextualizing agency-level performance.

International academic benchmark:

The MetroPT dataset (Porto Metro, Portugal, 2022), published by Veloso et al. [4], provides sensor-level data with ground-truth failure labels from a non-US transit system. This is the primary cross-system validation dataset — used to test

whether a model trained on MTA data generalizes to a structurally different transit network.

How we build training examples:

Each example is one piece of equipment over one 30-day window. If any unplanned outage was recorded for that unit in that window, $Y = 1$. Otherwise $Y = 0$. We slide that window forward one month at a time across the full historical record to maximize the number of labeled training examples available.

How We Know If It Works: Holdout Evaluation

We test the model using a holdout set — data from a time period the model has never seen. The principle is the same as training for a test using last year's exam questions, then sitting for this year's. The model studies on years one through eight, and we grade it on year nine.

- Training set: MTA outage data, January 2015 – December 2022
- Holdout (test) set: January 2023 – December 2024

The holdout set contains both X (equipment characteristics, maintenance logs, and weather for 2023–2024) and Y (the actual outages that happened in 2023–2024). The model makes predictions on the holdout data, we compare those predictions to what actually happened, and that comparison is the grade.

A note on why we do not use accuracy as our main metric:

In any 30-day window, the vast majority of equipment does not fail. If only 5% of units fail in a given month, a model that simply predicted “no failure” for everything would be 95% accurate — and completely useless. Accuracy hides the thing we actually care about: did we catch the equipment that was about to break? So we use three other measures instead.

How we measure performance:

AUC-ROC (discrimination ability)

If we show the model two pieces of equipment — one that will fail and one that will not — how often does it correctly rank the failing one as higher risk? A score of 1.0 means it gets it right every time. A score of 0.5 means it is guessing randomly. AUC measures this across every possible threshold setting, giving us a single number for overall model quality.

Precision and Recall (the trade-off)

Precision answers: of all the equipment we flagged as “at risk,” how many actually failed? High precision means fewer wasted maintenance dispatches.

Recall answers: of all the equipment that actually failed, how many did we catch? High recall means fewer Disabled riders get stranded.

These two measures pull in opposite directions. Flagging more equipment as at-risk catches more true failures (better recall) but also generates more false alarms (worse precision). The right balance depends on the agency's budget and legal risk tolerance. We publish the full trade-off curve — called a Precision-Recall Curve — so each agency can choose their own operating point rather than having us choose it for them.

F2-Score (our headline number)

The F2-Score combines precision and recall into one number, but weights recall twice as heavily as precision. The ‘2’ is the equity argument expressed as math: missing a real failure is twice as costly as dispatching an unnecessary maintenance call. Every modeling decision we make — the decision threshold, the class weights, the oversampling approach — is designed to push this number up.

The Two Ways the Model Can Be Wrong — and Why They Are Not Equal

Every prediction model makes two kinds of mistakes. Getting clear on the cost of each one drives every design decision in UpLift.

False Positive: We flag equipment as at-risk — it does not break

A maintenance crew goes out to service a unit that turns out to be fine. The cost is a wasted service call — somewhere between \$500 and \$2,000 in labor and materials. That is real money. But it is recoverable. No one is harmed.

False Negative: We miss a failure — equipment breaks without warning

A Disabled rider shows up at the station. The elevator is broken. They cannot reach the platform. They miss a medical appointment, a job interview, a family event. If this happens to the same riders repeatedly at the same stations, federal law treats it as a pattern of discrimination — not just a maintenance lapse. [6] ADA non-compliance settlements have reached nine figures. The human cost is not recoverable.

The gap between these two outcomes is not close. A false negative is categorically worse than a false positive.

Every technical decision in UpLift — the decision threshold, the class weights, the choice of F2 as our headline metric — reflects that asymmetry explicitly. We are willing to dispatch a few extra maintenance calls in order to miss fewer failures. That trade-off is a policy position, and we are making it consciously.

Specific measures to reduce false negatives:

- Lower decision threshold: We flag equipment as at-risk when the model's probability score reaches 0.3, not the default 0.5. This catches more true failures at the cost of more false alarms — a trade we accept.
- Class weighting: During training, the model is penalized more severely for missing a failure than for issuing a false alarm. This directly pushes the model toward recall.
- SMOTE oversampling: Because failures are rarer than non-failures in the training data, there is a risk the model learns to predict "no failure" for everything and scores well by default. SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic failure examples during training to prevent that from happening.
- Full Precision-Recall curve published: Rather than publishing a single point-in-time accuracy number, we publish the complete trade-off curve so agencies can select the threshold that fits their budget and legal exposure.

Making UpLift Work Everywhere

A tool that works only for the MTA is a tool for New York. A tool that works for every transit system is infrastructure for global disability equity. UpLift is designed for both, through four complementary strategies.

1. Standardized Feature Schema

The twelve input variables UpLift uses are universal. Equipment age, maintenance history, usage load, weather, inspection scores — these exist in every transit system on earth. We define a common data format that any agency can map their internal records to, regardless of what system they use to store them. This is the prerequisite for everything else.

2. Transfer Learning

We train a strong base model on the richest available dataset — MTA NYC, 2015 to 2024. Then we fine-tune that model using whatever local data a new agency has. Think of it like a specialist who has trained on thousands of cases in New York and then takes a posting in Nairobi. The core expertise transfers. The local context —

ridership patterns, climate, equipment mix — is learned on the job. An agency with only one year of maintenance records can still deploy UpLift, because the base model already understands the underlying physics of what aging equipment under heavy load tends to do.

3. Federated Learning

Many transit agencies will not share raw maintenance data. It may be legally restricted, politically sensitive, or simply proprietary. Federated learning works around this: each agency trains a local version of the model on their own data, then shares only the model's learned parameters — not the data itself — to improve a shared global model. The global model gets better without any agency exposing their records. This is the same mechanism used to improve smartphone keyboard predictions without the phone manufacturer ever reading your messages.

4. Multi-System Data Consortium

Over time, we build a coalition of transit agencies willing to contribute anonymized outage records in exchange for access to UpLift and its predictions. We start with agencies that already publish this data publicly: MTA NYC (data.ny.gov), WMATA DC, CTA Chicago, and the MetroPT dataset from Porto. Each new partner improves the model for everyone. The incentive structure is straightforward: share anonymized data, get a better prediction model back.

Risks and Mitigations

Risk	Mitigation
Underreported outages: MTA has been cited by auditors for failing to log all outages, which inflates uptime figures and means some failures in our training data are missing their Y label.	Cross-validate outage logs against rider complaint records (311 NYC), social media reports, and crowdsourced accessibility apps. Partner with disability advocacy organizations as independent validators.
Class imbalance: Most equipment does not fail in any given 30-day window. The model may learn to always predict 'no failure' and score well by default without actually being useful.	SMOTE oversampling, inverse-frequency class weighting, and F2-Score as the primary metric all address this directly. We also monitor failure-class performance separately, not just aggregate accuracy.
Equipment heterogeneity: A 1985 MTA elevator and a 2019 Schindler unit behave differently. Treating them identically in one model may reduce accuracy.	Equipment age, manufacturer, and model are explicit input features. We also explore training separate sub-models by equipment category and combining their outputs.
Domain shift: A model trained on New York infrastructure may not generalize to cities with different climates, ridership volumes, or maintenance cultures.	MetroPT (Porto) provides a cross-system validation test. Transfer learning and local fine-tuning address this structurally. We are explicit with partner agencies about where the model's confidence is lower.
Equity bias: If maintenance has historically been worse at stations in lower-income or majority-minority neighborhoods, the model may learn to deprioritize those stations — encoding and amplifying existing inequity.	Post-training equity audit: we test whether predicted maintenance priority correlates with neighborhood income or demographics. If it does, we apply fairness constraints. All agency deployment contracts require equity impact reporting.
Over-reliance: Agencies may reduce manual inspections because they trust the model, creating new blind spots.	UpLift is explicitly a decision-support tool, not a replacement for inspections. All outputs include uncertainty estimates. Contracts preserve minimum manual inspection schedules.

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